

Econometrics Methods I : The dynamics of Financial returns

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Contents

1	AI statement	2
2	Introduction	2
3	Data	2
4	Data Description	2
5	Testing for Autocorrelation and Heteroskedasticity	3
5.1	Autocorrelation	3
5.2	Heteroskedasticity	4
6	Robust Standard Error (HCSE)	5
6.1	Comparison with previous results	5
7	Forecasting	5
7.1	Forecasting Returns	5
7.2	Forecasting Volatilities	6
8	Granger Causality	9
9	Robust CAPM	9
10	Conclusion	10

1 AI statement

We used AI to built our R script to gather the data from Yahoo Finance. AI also helped to make the figures and tables in Latex. Finally, we used it for the commands of Eviews.

2 Introduction

This empirical project studies the dynamics of financial returns using daily stock price data. The main objective is to analyse the first and second moments of stock returns. The first moment refers to the predictability of returns, namely whether current returns can be explained by past returns. The second moment refers to the behaviour of volatility over time, in particular whether returns display time-varying volatility or volatility clustering.

The analysis is based on daily closing prices for four selected stocks and one market index over a period of more than ten years. The selected stocks are Apple Inc. (AAPL), Lockheed Martin (LMT), JPMorgan Chase & Co. (JPM), and Shell plc (SHEL). The market index used as a benchmark is the S&P 500 index, represented by the ticker `^GSPC`.

These assets were selected to cover large and internationally relevant companies from different sectors of the economy. Apple represents the technology sector, Lockheed Martin represents the defense sector, JPMorgan Chase represents the financial sector, and Shell represents the energy sector. The S&P 500 is included as a broad market index and will also be used later in the analysis for comparison and for the CAPM regression.

For each asset, the main variable of interest is the daily log-return, defined as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \quad (1)$$

where P_t denotes the closing price of the asset at day t . The empirical analysis will first describe the statistical properties of the return series, before testing for autocorrelation and heteroskedasticity. The project will then compare forecasting models for returns and volatility, examine possible Granger causality between the selected assets, and estimate a robust CAPM model using the market index.

3 Data

We gathered the data using the *tidyquant* library in R. We used the adjusted closing price, this is the closing price corrected for corporate actions such as dividends. The data is for 5 day workweeks from Monday to Friday. missing dates such as holidays have been added using the simple average.

$$\text{missing day}_t = \frac{P_{t-1} + P_{t+1}}{2}$$

we have data from January 2 2010 to 31 December 2024.

4 Data Description

Table 1 reports the mean, standard deviation, skewness, kurtosis, and the Jarque–Bera normality test statistic for the five log-return series $r_t = \ln(P_t) - \ln(P_{t-1})$.

Table 1: Descriptive Statistics for Daily Log>Returns (01/04/2010 – 12/31/2024)

Asset	Obs.	Mean	Median	Max	Min	Std. Dev.	Skew.	Kurt.	JB stat
AAPL	3911	0.000936	0.000969	0.1132	−0.1377	0.01708	−0.238	8.913	5734*
LMT	3911	0.000591	0.000734	0.1019	−0.1355	0.01285	−0.602	17.900	36413*
JPM	3911	0.000539	0.000521	0.1656	−0.1621	0.01701	−0.071	13.445	17782*
SHEL	3911	0.000200	0.000304	0.1795	−0.1884	0.01728	−0.804	20.406	49792*
^GSPC	3911	0.000421	0.000575	0.0896	−0.1277	0.01051	−0.764	17.143	32975*

All five return series reject the null hypothesis of normality at the 5% significance level. The Jarque–Bera statistics are far above the critical value of 5.99, with the smallest value being 5,734 for AAPL. All

series are negatively skewed, indicating a heavier left tail relative to a normal distribution. This means that large negative returns occur more frequently than a symmetric distribution would predict.

All series also display excess kurtosis, confirming the presence of fat tails that are typical of daily financial returns. The kurtosis is most extreme for SHEL (20.41) and LMT (17.90), suggesting that large daily moves are particularly common for these stocks. AAPL has the lowest kurtosis among the individual stocks (8.91), while the S&P 500 falls in between (17.14).

In terms of average returns, AAPL has the highest mean daily return (0.094%), consistent with its strong performance over the sample period. SHEL has the lowest mean daily return (0.020%). The S&P 500 has the lowest standard deviation (1.05%), which is expected since it is a diversified index. The individual stocks all have standard deviations of approximately 1.7%, reflecting their higher idiosyncratic risk.

5 Testing for Autocorrelation and Heteroskedasticity

5.1 Autocorrelation

Autocorrelation means that the error term in one period is correlated with the error term in a previous period. In this case, we test for first-order autocorrelation, meaning correlation between ε_t and ε_{t-1} .

The baseline return equation is:

$$r_t = \alpha + \varepsilon_t, \quad t = 1, \dots, T$$

To test whether the residuals are autocorrelated, we first take the residuals $\hat{\varepsilon}_t$ from the OLS regression and then estimate the following auxiliary regression:

$$\hat{\varepsilon}_t = \gamma + \rho \hat{\varepsilon}_{t-1} + u_t$$

The null and alternative hypotheses are:

$$H_0 : \rho = 0$$

$$H_1 : \rho \neq 0$$

Under the null hypothesis, there is no first-order autocorrelation. The test statistic is:

$$t = \frac{\hat{\rho} - 0}{se_{\hat{\rho}}}$$

At the 5% significance level, the null hypothesis of no autocorrelation is rejected when:

$$|t| > 1.96$$

Table 2: Autocorrelation Test Using t-statistics

Ticker	$\hat{\rho}$	$se_{\hat{\rho}}$	t-statistic	Decision	Conclusion
AAPL	-0.0298	0.0160	-1.8666	Do not reject H_0	No autocorrelation
LMT	-0.0058	0.0160	-0.3630	Do not reject H_0	No autocorrelation
S&P 500	-0.1048	0.0159	-6.5866	Reject H_0	Autocorrelation present
JPM	-0.0890	0.0159	-5.5886	Reject H_0	Autocorrelation present
SHEL	0.0065	0.0160	0.4092	Do not reject H_0	No autocorrelation

The results show that the absolute t-statistics for the S&P 500 and JPM are larger than 1.96. Therefore, we reject the null hypothesis of no first-order autocorrelation for these two series. This means that their residuals are significantly correlated with their own lagged values. For AAPL, LMT, and SHEL, the absolute t-statistics are smaller than 1.96, so we do not reject the null hypothesis. Therefore, there is no statistically significant evidence of first-order autocorrelation for these three return series.

As an additional test, we examine whether past returns can help explain current returns. This is done by estimating the following two equations:

$$r_t = \alpha + \varepsilon_t$$

$$r_t = \alpha + \beta r_{t-1} + \varepsilon_t$$

The key hypothesis is:

$$H_0 : \beta = 0$$

$$H_1 : \beta \neq 0$$

If $\beta = 0$, past returns do not help predict current returns. If the p-value of the lagged return coefficient is below 0.05, we reject H_0 and conclude that past returns contain statistically significant information about current returns.

Table 3: Return Predictability Test Using Lagged Returns

Ticker	Lagged return coefficient	t-statistic	p-value	Decision	Conclusion
AAPL	0.0176	0.0328	0.9739	Do not reject H_0	Not predictable
S&P 500	-0.5754	-3.7983	0.0001	Reject H_0	Predictable
LMT	-1.7036	-1.4348	0.1514	Do not reject H_0	Not predictable
SHEL	7.4648	3.0836	0.0021	Reject H_0	Predictable
JPM	-0.6937	-3.9589	0.0001	Reject H_0	Predictable

The return predictability results show that the lagged return coefficient is statistically significant for the S&P 500, SHEL, and JPM. For these series, the null hypothesis that $\beta = 0$ is rejected, suggesting that past returns have some predictive power for current returns. For AAPL and LMT, the p-values are above 0.05, so the null hypothesis is not rejected. Therefore, there is no statistically significant evidence that past returns predict current returns for these two stocks.

5.2 Heteroskedasticity

Heteroskedasticity means that the variance of the error term is not constant over time. In the context of financial returns, this is especially relevant because volatility often changes over time. To test for this, we use an ARCH test based on the squared residuals.

First, we estimate the following return equation by OLS:

$$r_t = \alpha + \varepsilon_t, \quad t = 1, \dots, T$$

We then take the residuals from this regression, $\hat{\varepsilon}_t$, and estimate an auxiliary regression using squared residuals:

$$\hat{\varepsilon}_t^2 = \mu + \phi \hat{\varepsilon}_{t-1}^2 + u_t$$

The squared residuals are used as a proxy for the conditional variance. This is similar to the autocorrelation test, but instead of testing whether residuals are correlated over time, we test whether squared residuals are correlated over time.

The null and alternative hypotheses are:

$$H_0 : \phi = 0$$

$$H_1 : \phi \neq 0$$

Under the null hypothesis, there are no ARCH effects, meaning that past volatility does not help explain current volatility. If the p-value is below 0.05, we reject H_0 and conclude that the return series shows evidence of heteroskedasticity.

The test statistic for the coefficient on the lagged squared residual is:

$$t = \frac{\hat{\phi} - 0}{se_{\hat{\phi}}}$$

For all five return series, the p-value of the t-test is equal to 0.0000. Therefore, we reject the null hypothesis of homoscedasticity for AAPL, the S&P 500, JPM, SHEL, and LMT. This means that there is strong evidence of heteroskedasticity. In practical terms, the variance of returns is not constant over time, and volatility tends to cluster across periods.

Table 4: ARCH Test for Heteroskedasticity Using t-statistics

Ticker	$\hat{\phi}$	$se_{\hat{\phi}}$	t-statistic	p-value	Conclusion
AAPL	0.456630	0.014231	32.08638	0.0000	Heteroskedasticity present
S&P 500	0.393137	0.014709	26.72849	0.0000	Heteroskedasticity present
JPM	0.227311	0.015578	14.59223	0.0000	Heteroskedasticity present
SHEL	0.208848	0.015644	13.35043	0.0000	Heteroskedasticity present
LMT	0.286964	0.015324	18.72677	0.0000	Heteroskedasticity present

6 Robust Standard Error (HCSE)

Table 5: Return Predictability Using HCSE Standard Errors

Asset	$\hat{\beta}$	Robust Std. Error	t-statistic	p-value	Adjusted R^2
AAPL	-0.029842	0.026393	-1.130694	0.2583	0.000635
LMT	-0.005805	0.038598	-0.150403	0.8805	-0.000222
JPM	-0.089031	0.036344	-2.449700	0.0143	0.007675
SHEL	0.006546	0.036036	0.181643	0.8559	-0.000213
GSPC	-0.104772	0.042481	-2.466337	0.0137	0.010724

Using HCSE standard errors, we test whether lagged returns help explain current returns. For AAPL, LMT and SHEL, the lagged return coefficient is not statistically significant, since the p-values are above 0.05. Therefore, we do not reject the null hypothesis that $\beta = 0$. For JPM and GSPC, the lagged return coefficient is statistically significant at the 5% level. In both cases, the coefficient is negative, suggesting short-term reversal in returns. However, the adjusted R^2 values remain very close to zero for all series. This means that even when lagged returns are statistically significant, they explain only a very small fraction of daily return movements.

6.1 Comparison with previous results

Table 6: Comparison Between OLS and HCSE Inference

Asset	$\hat{\beta}$	OLS t-statistic	HCSE t-statistic	OLS decision	HCSE decision	Change?
AAPL	-0.029842	-1.8666	-1.1307	Do not reject H_0	Do not reject H_0	No
LMT	-0.005805	-0.3630	-0.1504	Do not reject H_0	Do not reject H_0	No
JPM	-0.089031	-5.5886	-2.4497	Reject H_0	Reject H_0	No
SHEL	0.006546	0.4092	0.1816	Do not reject H_0	Do not reject H_0	No
GSPC	-0.104772	-6.5866	-2.4663	Reject H_0	Reject H_0	No

The comparison between the standard OLS inference (table 2) and the HCSE inference (table 5) shows that the estimated lagged return coefficients remain unchanged, while the t-statistics change after correcting the standard errors for heteroskedasticity. This is expected because HCSE only adjusts the estimated variance-covariance matrix, not the OLS coefficient estimates. For all five assets, the final decision remains unchanged. AAPL, LMT and SHEL remain statistically insignificant, while JPM and GSPC remain significant at the 5% level. However, the HCSE t-statistics for JPM and GSPC are much smaller in absolute value than the original OLS t-statistics, showing that standard OLS inference overstated the strength of the evidence when heteroskedasticity was ignored.

7 Forecasting

7.1 Forecasting Returns

Based on the HCSE results in the previous section, we select the forecasting model for each asset. For AAPL, LMT and SHEL, the lagged return coefficient was not statistically significant, so we use the simple mean return model:

$$r_t = \alpha + \varepsilon_t$$

For JPM and GSPC, the lagged return coefficient was statistically significant at the 5% level, so we use the autoregressive model with one lag:

$$r_t = \alpha + \beta r_{t-1} + \varepsilon_t$$

Following the instructions, each model is reestimated after excluding the last 500 observations. The estimated parameters are then kept fixed and used to produce static one-step-ahead forecasts. Forecast accuracy is evaluated using the root mean squared error:

$$RMSE = \sqrt{\frac{1}{500} \sum_{i=1}^{500} (\hat{r}_{T_1+i} - r_{T_1+i})^2}$$

Table 7: Static One-Step-Ahead Forecast Accuracy for Returns

Asset	Selected model	RMSE	Mean Absolute Error
AAPL	$r_t = \alpha + \varepsilon_t$	0.017601	0.012303
LMT	$r_t = \alpha + \varepsilon_t$	0.013133	0.008762
JPM	$r_t = \alpha + \beta r_{t-1} + \varepsilon_t$	0.017446	0.011832
SHEL	$r_t = \alpha + \varepsilon_t$	0.017912	0.011733
GSPC	$r_t = \alpha + \beta r_{t-1} + \varepsilon_t$	0.010868	0.007103

Table 7 reports the accuracy of the static one-step-ahead return forecasts. The S&P 500 has the lowest RMSE, equal to 0.010868, meaning that it has the smallest forecast errors among the five series. LMT also performs relatively well, with an RMSE of 0.013133. AAPL, JPM and SHEL have higher RMSE values, indicating less accurate return forecasts.

Overall, the forecasts remain very close to zero. This is consistent with the earlier results, where the adjusted R^2 values were close to zero. Even when the lagged return is statistically significant for JPM and GSPC, the forecasting power remains limited. This confirms that daily returns are difficult to predict using only their own past values.

7.2 Forecasting Volatilities

To model time-varying volatility, we estimate an ARCH(1) model for each return series. The mean equation is kept the same as in the return forecasting section. Therefore, AAPL, LMT and SHEL are estimated with an intercept-only mean equation, while JPM and GSPC include one lagged return because the lagged return coefficient was statistically significant in the HCSE analysis.

The ARCH(1) model is written as:

$$r_t = \alpha + \beta r_{t-1} + e_t$$

$$e_t = u_t \sqrt{h_t}, \quad u_t \sim N(0, 1)$$

$$h_t = \omega + \gamma e_{t-1}^2$$

For AAPL, LMT and SHEL, the coefficient β is omitted from the mean equation.

Table 8: ARCH(1) Volatility Model Estimates

Asset	Mean equation	$\hat{\omega}$	p-value	$\hat{\gamma}$	p-value
AAPL	$r_t = \alpha + e_t$	0.000232	0.0000	0.270203	0.0000
LMT	$r_t = \alpha + e_t$	0.000115	0.0000	0.320017	0.0000
JPM	$r_t = \alpha + \beta r_{t-1} + e_t$	0.000177	0.0000	0.418001	0.0000
SHEL	$r_t = \alpha + e_t$	0.000192	0.0000	0.437029	0.0000
GSPC	$r_t = \alpha + \beta r_{t-1} + e_t$	0.0000655	0.0000	0.505060	0.0000

Table 8 reports the ARCH(1) estimates. For all five assets, the coefficient on the lagged squared residual, $\hat{\gamma}$, is positive and statistically significant with a p-value of 0.0000. This means that past shocks have a significant effect on current volatility. In other words, the return series display volatility clustering.

The ARCH effect is strongest for GSPC, with $\hat{\gamma} = 0.505060$, followed by SHEL and JPM. This suggests that large shocks in these series are more likely to be followed by periods of higher volatility. AAPL has the lowest ARCH coefficient, but it is still statistically significant.

Table 9: Comparison of Return Forecast RMSE With and Without ARCH

Asset	RMSE without ARCH	RMSE with ARCH	Difference
AAPL	0.017601	0.017613	0.000012
LMT	0.013133	0.013179	0.000046
JPM	0.017446	0.017491	0.000045
SHEL	0.017912	0.017982	0.000070
GSPC	0.010868	0.010920	0.000052

Table 9 compares the return forecast accuracy of the standard forecasting models with the ARCH(1) models. For all five assets, the RMSE with ARCH is slightly higher than the RMSE without ARCH. Therefore, the ARCH(1) model does not improve the point forecasts of returns in this sample.

However, this does not mean that the ARCH model is useless. Its main contribution is to model time-varying volatility and produce forecast intervals that change over time. The ARCH model captures volatility clustering, meaning that periods of large shocks are followed by periods of higher forecasted variance. Therefore, ARCH improves the modelling of uncertainty, even if it does not improve the accuracy of the return point forecasts.

The plots in figure 1 confirm the presence of volatility clustering for all five assets. Periods with large return shocks are followed by spikes in the forecasted variance, especially around 2020, showing that the ARCH(1) model captures changes in uncertainty over time.

The return forecasts remain close to zero and do not track daily returns very well. This supports the RMSE results: the ARCH(1) model is useful for modelling time-varying volatility, but it does not improve the accuracy of return point forecasts.

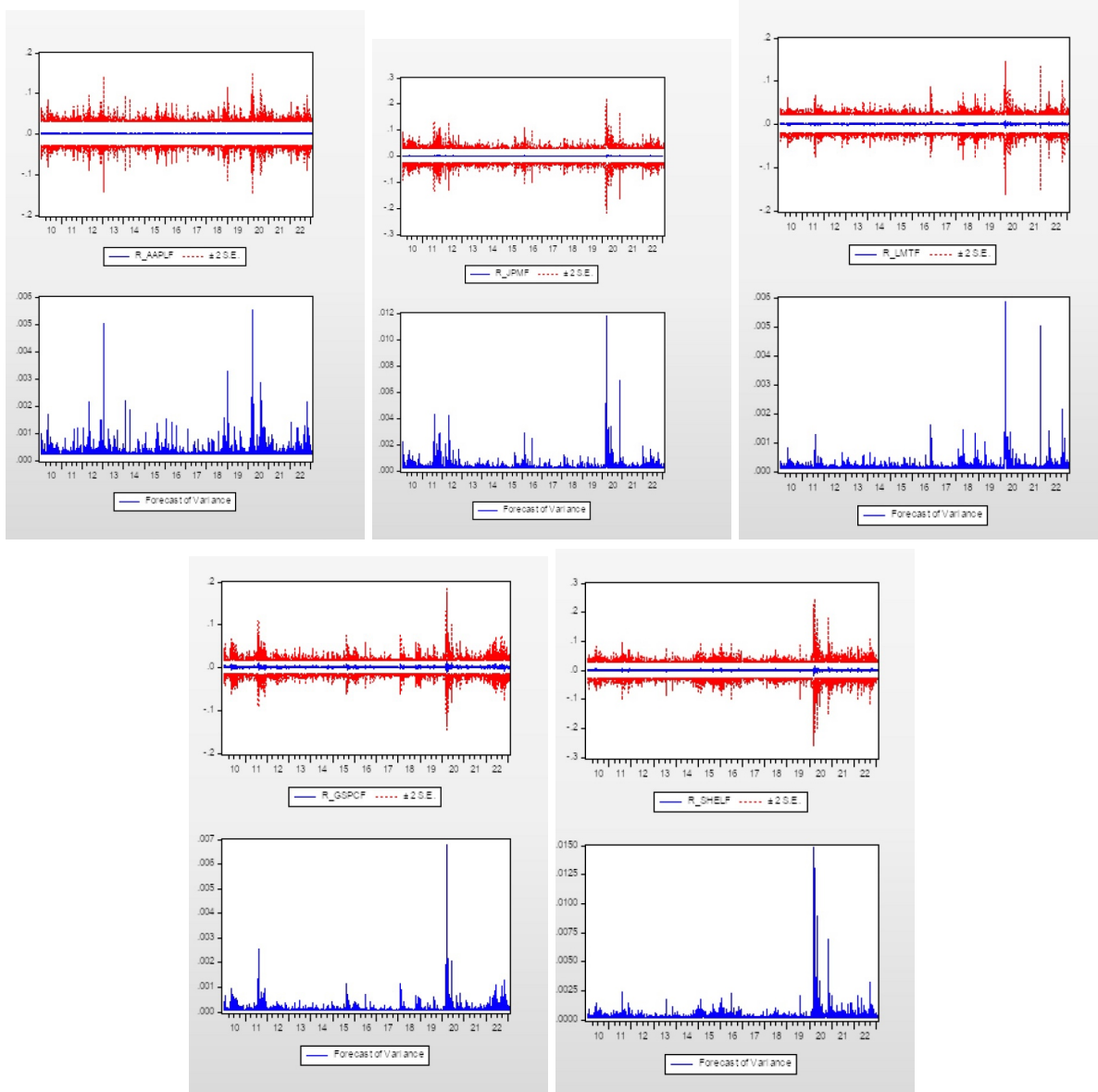


Figure 1: ARCH(1) Return Forecasts and Forecasted Variances for All Assets

8 Granger Causality

In this section, we test whether the lagged returns of the other stocks help predict the current return of each asset. Following the instructions, we estimate each equation separately by OLS using HCSE robust standard errors. The general model is:

$$r_{i,t} = \alpha_i + \beta_{i,AAPL}r_{AAPL,t-1} + \beta_{i,JPM}r_{JPM,t-1} + \beta_{i,LMT}r_{LMT,t-1} + \beta_{i,SHEL}r_{SHEL,t-1} + \varepsilon_{i,t}$$

For each equation, we test whether the lagged returns of the other assets are statistically significant. If the p-value is below 0.05, we reject the null hypothesis of no Granger causality.

Table 10: Granger Causality Regressions Using HCSE Standard Errors

Dep. var.	Sig. predictors	AAPL	JPM	LMT	SHEL	Adj. R^2	Conclusion
AAPL	None	—	0.0960	0.3337	0.9281	0.005398	No Granger causality
JPM	AAPL	0.0396	—	0.0866	0.3607	0.013450	AAPL → JPM
LMT	None	0.1076	0.2766	—	0.5175	0.002419	No Granger causality
SHEL	AAPL	0.0039	0.0760	0.0840	—	0.018214	AAPL → SHEL

Table 10 reports the Granger causality regressions estimated with HCSE standard errors. For AAPL, none of the lagged returns of the other assets are statistically significant at the 5% level. Therefore, we do not find evidence that JPM, LMT or SHEL Granger-cause AAPL.

For JPM, the lagged return of AAPL is statistically significant with a p-value of 0.0396. This suggests that AAPL contains some predictive information for JPM returns. However, the lagged returns of LMT and SHEL are not significant at the 5% level.

For LMT, none of the lagged returns of the other assets are statistically significant. Therefore, there is no clear evidence of Granger causality toward LMT.

For SHEL, the lagged return of AAPL is statistically significant with a p-value of 0.0039. This suggests that AAPL helps predict SHEL returns. JPM and LMT are not significant at the 5% level, although their p-values are relatively close to 0.05.

Overall, the evidence of Granger causality is limited. AAPL appears to have some predictive power for JPM and SHEL, but the adjusted R^2 values remain very small in all equations. This means that even when some lagged returns are statistically significant, the explanatory power of the models is weak.

9 Robust CAPM

The CAPM regression estimates the relationship between each stock return and the market return. The model is:

$$r_{i,t} = \alpha_i + \beta_i r_{m,t} + \varepsilon_{i,t}$$

where $r_{i,t}$ is the return of stock i , and $r_{m,t}$ is the return of the S&P 500. The coefficient β_i measures the sensitivity of each stock to market movements. A beta above one means that the stock tends to move more than the market, while a beta below one means that it is less sensitive to the market. The intercept α_i measures the average return not explained by the market.

Table 11: CAPM Estimates Using HCSE Standard Errors

Asset	$\hat{\alpha}$	$\hat{\beta}$	HCSE $se(\hat{\alpha})$	HCSE $se(\hat{\beta})$	Adj. R^2	β p-value
AAPL	0.000492	1.108694	0.000218	0.022767	0.477001	0.0000
JPM	-0.000008	1.211911	0.000193	0.029019	0.580212	0.0000
LMT	0.000390	0.687945	0.000186	0.033821	0.329778	0.0000
SHEL	-0.000183	0.985611	0.000248	0.051888	0.363932	0.0000

Table 11 shows that the market return is statistically significant for all four stocks. The p-values of the beta coefficients are equal to 0.0000, so we reject the null hypothesis that $\beta_i = 0$ for AAPL, JPM, LMT and SHEL. This means that all four stock returns are strongly related to movements in the S&P 500.

JPM has the highest beta, equal to 1.211911. This means that JPM is the most sensitive stock to market movements in our sample. If the S&P 500 increases by 1 percentage point, JPM is expected to increase by about 1.21 percentage points, on average. AAPL also has a beta above one, equal to 1.108694, so it is also more volatile than the market.

LMT has the lowest beta, equal to 0.687945. This suggests that Lockheed Martin is less sensitive to market movements than the other stocks. SHEL has a beta close to one, equal to 0.985611, meaning that its return moves approximately one-for-one with the market.

The adjusted R^2 values show how much of each stock's return variation is explained by the market return. The CAPM explains the largest fraction of JPM returns, with an adjusted R^2 of 0.580212. It also explains a large part of AAPL returns, with an adjusted R^2 of 0.477001. The explanatory power is lower for LMT and SHEL, with adjusted R^2 values of 0.329778 and 0.363932 respectively.

Table 12: 95% Confidence Intervals for CAPM Coefficients

Asset	OLS CI for α	HCSE CI for α	OLS CI for β	HCSE CI for β
AAPL	[0.000065, 0.000919]	[0.000065, 0.000919]	[1.069735, 1.147653]	[1.064071, 1.153317]
JPM	[-0.000389, 0.000372]	[-0.000387, 0.000370]	[1.177315, 1.246507]	[1.155034, 1.268788]
LMT	[0.000029, 0.000751]	[0.000025, 0.000755]	[0.655037, 0.720853]	[0.621656, 0.754234]
SHEL	[-0.000663, 0.000297]	[-0.000669, 0.000303]	[0.941887, 1.029335]	[0.883911, 1.087311]

Table 12 compares the 95% confidence intervals obtained using standard OLS standard errors and HCSE standard errors. The coefficient estimates are the same in both cases, but the confidence intervals differ because the standard errors are different.

For all four assets, the 95% confidence intervals for β do not include zero, both with OLS and HCSE standard errors. This confirms that the market return is a statistically significant predictor of each stock return. The HCSE confidence intervals for β are wider than the OLS intervals, especially for JPM, LMT and SHEL. This means that ignoring heteroskedasticity would underestimate the uncertainty around the beta estimates.

For α , the interpretation is different. The confidence intervals for JPM and SHEL include zero, meaning that their alphas are not statistically different from zero. For AAPL and LMT, the confidence intervals do not include zero, suggesting positive abnormal returns relative to the CAPM. However, these alpha values are very small in daily return terms, so the economic importance is limited.

Overall, the CAPM results show that market movements explain a meaningful part of stock return variation, especially for JPM and AAPL. Using HCSE standard errors does not change the estimated coefficients, but it gives more reliable confidence intervals in the presence of heteroskedasticity.

10 Conclusion

This project analysed the dynamics of daily financial returns for AAPL, LMT, JPM, SHEL and the S&P 500. The results show limited evidence of return predictability. JPM and the S&P 500 show statistically significant autocorrelation and significant lagged return coefficients when using HCSE standard errors, but the adjusted R^2 values remain close to zero. This means that even when past returns are statistically significant, they explain only a very small part of daily return movements.

The results are stronger for volatility. The ARCH tests reject homoscedasticity for all five return series, and the ARCH(1) estimates confirm significant volatility clustering. However, the ARCH(1) models do not improve the RMSE of return point forecasts compared with the simpler models. Their value is mainly in modelling time-varying uncertainty rather than improving the prediction of average returns.

The Granger causality results show only limited cross-asset predictability, with AAPL helping to predict JPM and SHEL. The CAPM results are more conclusive: all four stocks have statistically significant exposure to the S&P 500. JPM and AAPL have betas above one, meaning they are more sensitive to market movements, while LMT has the lowest beta. Overall, the main conclusion is that daily returns are difficult to forecast, but volatility and market exposure are clearly present and economically meaningful.